OCIMF

Oil Companies
International
Marine Forum

RECOMMENDATIONS FOR OIL TANKER MANIFOLDS AND ASSOCIATED EQUIPMENT

Fourth Edition 1991



OCIMF

Recommendations for Oil Tanker Manifolds and Associated Equipment

(Fourth Edition - August 1991)

Oil Companies International Marine Forum

Issued by the
Oil Companies International Marine Forum
First Published 1971
Second Edition 1975
Third Edition 1981
Fourth Edition 1991

ISBN 1 85609 017 5

The Oil Companies International Marine Forum (OCIMF) is a voluntary association of oil companies having an interest in the shipment and terminalling of crude oil and oil products. OCIMF is organised to represent its membership before, and consult with, the International Maritime Organization and other government bodies on matters relating to the shipment and terminalling of crude oil and oil products, including marine pollution and safety.

Notice of Terms of Use

While the advice given in this guide has been developed using the best information currently available, it is intended purely as guidance and to be used at the user's own risk. No responsibility is accepted by the Oil Companies International Marine Forum, or by any person, firm, corporation or organisation who or which has been in any way concerned with the furnishing of information or data, the compilation, publication, or any translation, supply or sale of this guide, for the accuracy of any information or advice given herein or for any omission herefrom or for any consequences whatsoever resulting directly or indirectly from compliance with or adoption of guidance contained herein even if caused by a failure to exercise reasonable care.

Printed & Published by
WITHERBY & CO. LTD
32/36 Aylesbury Street,
London EC1R 0ET, England

Introduction

The Oil Companies International Marine Forum first published "Standards for Oil Tanker Manifolds and Associated Equipment" in 1971. The guidance was subsequently revised in a Second Edition, published in 1975, primarily to take account of changes in practice reflecting the increased usage of hard arms, the introduction of large diameter floating hoses at some SBM's and the increased frequency of lightening operations.

A Third Edition, published in 1981, contained revised and expanded recommendations relating to lifting equipment.

In this Fourth Edition, no substantive changes have been made to the previously-published guidance relating to cargo and bunker manifolds and associated equipment. However, in order to take account of the requirement at a limited number of terminals for tankers in certain trades to transfer cargo vapours ashore, a new Annex has been included containing recommendations for vapour manifolds. It is stressed that these Annexed recommendations are provided purely for the guidance of those operators whose ships may trade to terminals required to install vapour collection systems. The recommendations supplement the uniform safety design standards for marine vapour recovery systems being developed by the International Maritime Organization (IMO) and the regulations for marine vapour control systems promulgated by the United States Coast Guard in 1990. The recommendations relate only to the provision of a uniform manifold arrangement and do not address other issues associated with vapour transfer systems such as the need to provide safeguards to prevent or control vapour release or to prevent fire and explosion.

Contents

Section		Page
1	Ship Tonnage Categories	1
2	Manifold Position	1
	2.1 Centre of Cargo Manifold	
	2.2 Distance from Ship's Side	
	2.3 Height of Cargo Manifold	
	2.4 Spacing	
	2.5 Number	
3	Spill Tank and Working Platform	2
	3.1 Size and Position	
	3.2 Spill Tank Draining	
	3.3 Strength of Working Platform	
	3.4 Design of Working Platform	
	3.5 Access to Working Platform	
	3.6 Combined Working Platforms and Spill Tanks	
4	Cargo Manifold : General	3
	4.1 Material	
	4.2 Strength	
	4.3 Marking of Manifolds	
	4.4 Loading Arm Jacks	
5	Cargo Manifold Specification and Fittings	4
	5.1 Valves	
	5.2 Distance Pieces	
	5.3 Reducers	
	5.4 Blank Flanges	
6	Deck Fittings	6
	6.1 Cruciform Bollards	
	6.2 Bollards (Bitts)	
	6.3 Fairleads	
	6.4 Deck Rings or Lugs	
7	Lightening Arrangements	7
	7.1 Fairleads	
	7.2 Bollards *(Bitts)*	
	7.3 Other Mooring Fairleads	
8	Hose Support at Ship's Side	7
	8.1 Curved Plate or Pipe Hose Support	
	8.2 Hose Support Position	
	8.3 Hose Support Strength	
9	Bunker Connections	8
	9.1 Number and Position	
	9.2 Spacing	
	9.3 Strength	
	9.4 General	
	9.5 Presentation Flanges	
	9.6 Other Connections	
10	Lighting	10
11	Lifting Equipment	10
	11.1 Safe Working Loads	
	11.2 Area of Operation of Lifting Equipment	
	11.3 Operating Criteria for Lifting Equipment	
	11.4 Other Criteria for Derrick Systems	
	11.5 Conflict of Criteria	

Annex

Recommendations for Vapour Manifolds

Section		Page
A.1	General	17
A.2	Manifold Position A.2.1 Spacing A.2.2 Number and Position A.2.3 Labelling	17
A.3	Spill Tank and Working Platform	18
A.4	Vapour Manifold : General A.4.1 Material and Strength	18
A.5	Vapour Manifold: Specification and Fittings A.5.1. Valves and Distance Pieces A.5.2 Reducers/Expanders	18

List of Tables

Table		Page
1	Minimum Cargo Manifold Spacing	1
2	Loads at Presentation Flange	3
3	Cargo Reducers	5
4	Working Load of Bollards	6
5	Hose Support Curvature	7
6	Minimum Bunker Line Spacing	8
7	Loads at Bunker Presentation Flange	8
8	Bunker Reducers	9
9	Lifting Equipment Safe Working Load	10
A.1	Minimum Vapour Manifold Spacing	17
A.2	Vapour Reducers/Expanders	18

List of Figures

Figure		Page
1	Category A. Standard Manifold Arrangement for 16,000 to 25,000 DWT Tankers	12
2	Category B. Standard Manifold Arrangement for 25,001 to 60,000 DWT Tankers	13
3	Category C. Standard Manifold Arrangement for 60,001 to 160,000 DWT Tankers	14
4	Category D. Standard Manifold Arrangement forTankers over 160,000 DWT	15
5	Loading Arm Jack Bearing Surfaces	16
6	Loading Arm Loads (Definition of Forces)	16
7	Presentation Flange Orientation	16
A.1	Category A. Standard Manifold Arrangement for 16,000 to 25,000 DWT Tankers including Vapour Lines	20
A.2	Category B. Standard Manifold Arrangement for 25,001 to 60,000 DWT Tankers including Vapour Lines	21
A.3	Category C. Standard Manifold Arrangement for 60,001 to 160,000 DWT Tankers including Vapour Lines	22
A.4	Category D. Standard Manifold Arrangement for Tankers over 160,000 DWT including Vapour Lines	23
A.5	Vapour Manifold Presentation Flange Orientation	24
A.6	Labelling of Vapour Manifolds	24

1. SHIP TONNAGE CATEGORIES

A	16,000 to 25,000	SDWT tonnes (Fig. 1 refers)
B	25,001 to 60,000	SDWT tonnes (Fig. 2 refers)
C	60,001 to 160,000	SDWT tonnes (Fig. 3 refers)
D	Ships over 160,000	SDWT tonnes (Fig. 4 refers)

2. MANIFOLD POSITION

2.1
Centre of Cargo Manifold

The centre of the cargo manifold arrangement should be located at the mid-length of the ship, or as near as possible, but in no case should it be more than 3.0m forward or aft of the mid-length. For these purposes, the ship's length should be taken as the length overall.

2.2
Distance from Ship's Side

The distance of the presentation flanges inboard from the ship's side should be 4.6m.

2.3
Height of Cargo Manifold

2.3.1 The centres of the manifold presentation flanges should be located at least 700mm above the horizontal projection of the top of the hose support at the ship's side. (See Section 8).

2.3.2 The height of the centres of the presentation flanges above the deck should not exceed 2.1m.

2.3.3 The working platform should be fitted to allow 900mm between the level of the platform and the centres of the presentation flanges. (See Section 3.)

2.4
Spacing

The minimum spacing of the cargo manifold as measured centre to centre along the line of presentation flanges should be in accordance with Table 1 below.

TABLE 1 : MINIMUM CARGO MANIFOLD SPACING

Ship Tonnage Category	Cargo Manifold Spacing Centre to Centre
A	1.5 metres
B	2.0 metres
C	2.5 metres
D	3.0 metres

2.5
Number

Similar manifold arrangements should be provided on each side of the ship.

3. SPILL TANK AND WORKING PLATFORM

All ships should be provided with permanently fitted working platforms and spill tanks under the cargo and bunker manifold connections.

3.1
Size and Position

Length — The spill tank and the working platform should extend beyond the bunker connections at the forward and aft ends on the manifold. On ships provided with vapour return facilities (see Annex), the spill tank and the working platform should extend beyond the vapour manifold connections.

Width — The spill tank and working platform should have a width of approximately 1.8m and should be so positioned that about 1.2m is outboard of the reducer presentation flanges.

Depth – The spill tank should have a minimum depth of about 300mm.

Vertical Positioning – The working platform should be situated 900mm below the level of the centres of the reducer presentation flanges and the spill tank should be located below the working platform.

3.2
Spill Tank Draining

Suitable means of draining the spill tank should be provided. When considering draining facilities, the particular type of tanker, its trading pattern and the cargoes likely to be carried, should be taken into account.

3.3
Strength of Working Platform

The strength and construction of the working platform and supports should be such that all normal loads encountered during hose connection, cargo handling and disconnection can be adequately catered for. The grids forming the working surface should be capable of withstanding a load of at least 1 tonne/m^2. (See also Section 4.4).

3.4
Design of Working Platform

The working platform surface should be made from open mesh grating to allow free passage of any oil spilt from the manifold into the spill tank.

The grid should have a non-slip finish and should be removable, in easily handled sections, to permit access to the spill tank.

The outboard uppermost edge of the working platform/spill tank should be fitted with a 50mm radius rounded edge to protect cargo and bunker hoses during connection and disconnection.

3.5
Access to Working Platform

A steel stairway of suitable size should be provided at the forward and aftermost ends of the platform leading to the deck.

3.6
Combined Working Platforms and Spill Tanks

Where the working platform and spill tank have been designed as a combined unit, the dimensions and strength should conform with those outlined in 3.1 and 3.3. In addition, the other provisions in Section 3 should be followed as closely as possible, including those regarding the provision of hose protection on leading edges.

4. CARGO MANIFOLD:GENERAL

4.1
Material

The whole cargo manifold, including the manifold support, should be of steel and should preferably conform to the requirements of ANSI B31.3 - Chemical Plant and Petroleum Refinery Piping, 1987.

4.2
Strength

The cargo manifold supports, distance pieces and reducers should be capable of withstanding the loads at the presentation flange as outlined in Table 2 below. (See also Figure 6).

TABLE 2 : LOADS AT PRESENTATION FLANGE

	Tonnage Category/ Loads		
	A	B	C D
Vertical Force (tonnes)	5	10	15
Lateral Force (tonnes)	2	3	4
Axial Force (tonnes)	5	7	10
Moment (tonne metres)	5	10	15
Forces based on maximum loads imposed by unsupported loading arms of stated size	16"	20"	24"

4.3
Marking of Manifolds

All manifolds should be permanently and clearly marked with the loads that they have been designed to withstand.

4.4
Loading Arm Jacks

In all normal circumstances, manifolds constructed according to these recommendations should obviate the need for using loading arm jacks. Therefore, it should only be necessary to build working platforms to the strength parameters contained in Section 3.3.

If loading arm jacks are used, it will be necessary to provide additional stiffening or temporary means to distribute the forces by using timber or other suitable material as pads for the base of the loading arm jacks. The particular areas, over the working platform and deck, which are most likely to accept these occasional jack loads are shown in Figure 5. In order to guard against possible damage to a working platform from this cause, owners may wish to permanently strengthen the platform. In this case, the area outlined in Figure 5 should be capable of withstanding 100 tonnes per square metre.

5. CARGO MANIFOLD SPECIFICATION AND FITTINGS

5.1
Valves

5.1.1 *Material*

The cargo manifold valves should be made of steel and be fitted with flanges conforming preferably to B.S. 1560 or ANSI B16.5.

5.1.2 *Actuators*

Cargo manifold valve actuators, when fitted, should be placed on the inboard side of the valve to avoid damage and overcrowding during hose handling operations.

5.2
Distance Pieces

5.2.1 *Material*

Steel distance pieces with outboard flanges conforming to B.S.1560 or ANSI B16.5 should be fitted outboard of the manifold valves and immediately inboard of the reducer. The inboard flanges should preferably conform to B.S. 1560 or ANSI B16.5.

5.2.2 *Length*

The length of the distance pieces should be kept to a minimum. However, their length should be adequate to provide sufficient room for the following:-

i) To ensure that sufficient space is available for bolt withdrawal from the valve/distance piece and distance piece/reducer flanges. A minimum of 200mm should be allowed between each flange and the manifold support structure.

ii) To accommodate the manifold support.

iii) To accommodate other line connections including stripping discharges and drain lines which may be required by owners. Such lines, when incorporated, should be fitted inboard of the manifold supports.

5.3
Reducers

5.3.1 *Material*

All reducers should be made of steel and fitted with flanges that conform to B.S. 1560 or ANSI B16.5. The inboard flange should be made to bolt directly to the distance piece (see Section 5.2).

The outboard flange on all reducers should be known as the "presentation flange".

5.3.2 *Length*

All reducers should have an overall length of 500mm in order to allow sufficient space for the operation of quick acting couplers.

5.3.3 *Lifting Lugs*

All reducers should be fitted with a lifting lug. This lug should be placed as near to tne centre of balance as possible and at a location which should not interfere with either the operation of quick acting couplers or with the bolting up of flanges.

5.3.4 *Presentation Flanges*

The number of reducers carried and the size of presentation flanges should be in accordance with Table3. In normal circumstances, the principal cargo reducers should be kept bolted to the distance pieces. On the occasions when a smaller presentation flange is required, the semi-permanent principal cargo reducers should be removed and replaced by the required size reducers from the onboard stock.

5.3.5 *Material and Design*

All the presentation flanges shall be kept vertical and have plain (flat) faces, i.e. raised faced flanges should not be used. Gasket contact surfaces should be machined and finished with a continuous spiral groove in accordance with B.S. 1560 or ANSI B16.5

5.3.6 *Type of Flanges*

The presentation flanges should be "welded neck" flanges in order to provide a clear inner diameter for the insertion of alignment probes. If "slip on" flanges are fitted, the internal weld should be ground flush to conform to the nominal bore specification outlined in Table 3. Flange bolt holes should be aligned as shown in Figure 7.

TABLE 3: CARGO REDUCERS

	Principal Fittings		Reserve Reducer Stocks					
Ship Tonnage Category	Number of Reducers	Presentation Flange Nominal Bore	Number of Reducers	Presentation Flange Nominal Bore	Number of Reducers	Presentation Flange Nominal Bore	Number of Reducers	Presentation Flange Nominal Bore
A	8	12"	4	8"	4	10"	–	–
B	8	16"	4	8"	4	10"	4	12"
C	8	16"	4	8"	4	10"	4	12"
D	8	20"	4	12"	4	16"	–	–

NOTE: The numbers of principal and reserve reducers quoted above are based on the assumption that four manifold connections are provided on each side.

5.4
Blank Flanges

Each presentation flange of the principal reducers should be provided with a removable steel blank flange fitted with handles.

6. DECK FITTINGS

This section covers the design of the deck fittings which should be provided for hoisting and hanging off hoses at sea berths.

6.1
Cruciform Bollards

Sets of cruciform bollards, 600mm in height, should be welded to the deck on each side of the ship. The bollards should be fitted mid-way between each pair of cargo manifolds, allowing maximum clear area between the working platform and ship's side and yet giving sufficient room for safe access (about 300mm) between bollard and spill tank. The working load of these bollards should be as detailed in Table 4.

TABLE 4 : WORKING LOAD OF BOLLARDS

Ship Tonnage Category	Working Load
A B C	25 tonnes
D	40 tonnes

6.2
Bollards (Bitts)

Two pairs of bollards, 300mm in diameter for tonnage category A, B and C ships and 350mm in diameter for tonnage category D ships, should be provided on each side of the ship, one pair forward and one pair aft of the manifold.

These bollards should be conveniently placed near to, but outside of, the deck area between the manifold and the ship's side and in the same longitudinal line as the cruciform bollards (see Section 6.1).

The sets of bollards should be aligned and positioned as shown in Figures 1-4 and the working load should conform to Table 4.

6.3
Fairleads

Fairleads with clear openings 400 x 250 mm should be provided at the ship's side for hose chains and hose hoisting as detailed below:

a) For ship tonnage categories A and B - two fairleads having a working load of 25 tonnes should be provided on each side directly in line with the forward and aftermost cargo presentation flanges.

b) For ship tonnage categories C and D - fairleads having a working load of 40 tonnes should be provided on each side directly in line with each cargo manifold presentation flange.

For ships in category D, a clear run should be provided along the ship's deck to deck winches for leading the hose hoisting wires which are occasionally used at some sea terminals.

6.4
Deck Rings or Lugs

Deck rings or lugs of 15 tonnes working load should be fitted on each side of the ship in the following positions:-

i) On the deck adjacent to each bunker manifold and in the same longitudinal line as the cruciform bollards (see Section 6.1)

ii) On the deck adjacent to the bollards described in 6.2 (see plans, Figures 1 - 4).

iii) On the deck adjacent to each fairlead described in 6.3.

iv) On the deck adjacent to the hose support at the ship's side opposite each bunker connection.

7. LIGHTENING ARRANGEMENTS

Note: these recommendations are only applicable to ships within tonnage category D.

7.1

Fairleads

Two additional sets of heavy duty closed fairleads with clear openings of 500 x 400 mm should be positioned on the starboard side of the ship. These fairleads should be located 35m forward and aft of the mid-length of the ship or as close to this position as possible. All fairleads associated with ship to ship transfer operations should be smoothly finished on both inboard and outboard faces to prevent chafing of ropes.

7.2

Bollards (Bitts)

Twin sets of bollards of 550mm diameter should be located inboard of the fairleads (section 7.1) for handling the moorings used during lightening operations.

7.3

Other Mooring Fairleads

All other fairleads used during mooring should be of the closed type to avoid difficulties occurring owing to the large relative changes of freeboard experienced during lightening operations.

8. HOSE SUPPORT AT SHIP'S SIDE

Means should be provided to adequately support hoses in way of the ship's side abreast of the manifolds.

8.1

Curved Plate or Pipe Hose Support

In order to protect the hose from sharp edges and obstructions, a horizontal curved plate or pipe section should be fitted at the ship's side and above the fairleads described in Section 6.3.

The curved plate or pipe section should be fitted with its rounded surface uppermost and in such a position that the maximum surface area is presented to a hose coming on board from the sea.

The radius of curvature of this plate or pipe section should not be less than that quoted in Table 5 below.

TABLE 5 : HOSE SUPPORT CURVATURE

Ship Tonnage Category	Radius of Curvature
A B C	150 mm
D	300 mm

8.2

Hose Support Position

The hose supports should lie in the fore and aft axis at the ship's sides and be at least 700mm below the level of the centres of the presentation flanges. The hose support should extend beyond the centre of the bunker connections or, if applicable, vapour connections (see Annex), at the forward and aft ends of the manifold.

8.3

Hose Support Strength

a) For ship tonnage categories A and B - hose supports should be sized to cater for a minimum vertical load of two hoses each of 10 tonnes.

b) For ship tonnage category C - hose supports should be sized to cater for a minimum vertical load of two hoses each of 20 tonnes.

c) For ship tonnage category D - hose supports should be sized to cater for a minimum vertical load of two hoses each of 25 tonnes.

9. BUNKER CONNECTIONS

9.1
Number and Position

Four bunker connections should be provided, two on each side of the ship, with presentation flanges 4.6m from the ship's side and at the same height above the deck as the cargo manifold. In order that common pipe support arrangements can be utilised, it is acceptable for the base of the bunker connection to be at the same height as the base of the cargo connections.

One bunker connection should be located forward of the manifold and one located aft of the manifold on each side of the ship.

9.2
Spacing

The minimum spacing between the bunker presentation flanges and the nearest cargo manifold, as measured from centre to centre, should be in accordance with Table 6 below.

TABLE 6 : MINIMUM BUNKER LINE SPACING

Ship Tonnage Category	Spacing
A	1.5 metres
B C D	2.0 metres

9.3
Strength

The bunker manifold supports, distance pieces and reducers should be capable of withstanding the loads as outlined in Table 7. (See also Figure 6).

TABLE 7 : LOADS AT BUNKER PRESENTATION FLANGE

	Tonnage Category/Loads	
	A B	C D
Vertical Force (tonnes)	3	5
Lateral Force (tonnes)	1	2
Axial Force (tonnes)	2	5
Moment (tonne metres)	3	5
Forces based on maximum loads imposed by unsupported loading arms of stated size	12"	16"

9.4
General

The recommendations for the cargo manifolds contained in Section 5 should also be followed in the bunker manifold design except the length of reducers may, where appropriate, be reduced to 300mm.

9.5
Presentation Flanges

9.5.1 *Number and Size*

The number of reducers and the size of bunker presentation flanges for the various tonnage categories should be in accordance with Table 8.

In normal circumstances and depending upon the particular trade, the principal bunker reducer, as detailed in Table 8, should be kept bolted to the distance pieces. On the occasions when a smaller presentation flange is required, the semi-permanent principal reducers should be removed and replaced by the required size reducers from the onboard stock.

9.5.2 *Material, Design and Type of Flanges*

The material, design and type of flanges should conform with the recommendations contained in Sections 5.3.5 and 5.3.6.

TABLE 8: BUNKER REDUCERS

	Principal Fittings		Reserve Reducer Stocks					
Ship Tonnage Gategory	Number of Reducers	Presentation Flange Nominal Bore	Number of Reducers	Presentation Flange Nominal Bore	Number of Reducers	Presentation Flange Nominal Bore	Number of Reducers	Presentation Flange Nominal Bore
A	4	6"	1	8"	1	10"	–	–
B	4	6"	1	8"	1	10"	–	–
C	4	8"	1	10"	1	6"	1	12"
D	4	8"	1	10"	1	6"	1	12"

9.6
Other Connections

It is desirable that filling connections for Diesel Oil, Lube Oil and Fresh Water should be located in close proximity to the cargo and bunker manifolds on both the port and starboard sides.

10. LIGHTING

The levels of illumination in way of the manifold should not be less than 50 lux when measured on a parallel with and 1m above the deck.

This illuminated area should extend overside to the waterline in light displacement in order to provide illumination for handling hoses at sea berths.

Levels of illumination over the rest of the working deck should have an average minimum lighting intensity of 10 lux when measured on a parallel with and 1m above the deck.

11. LIFTING EQUIPMENT

11.1
Safe Working Loads

Ships should be provided with lifting equipment having safe working loads as detailed in Table 9. The safe working load of the lifting gear should be established in accordance with the requirements of the ship classification societies and should include a normal allowance for impact load.

TABLE 9 : LIFTING EQUIPMENT SAFE WORKING LOAD

Ship Tonnage Category	Minimum SWL
A B	10 tonnes
C	15 tonnes
D	20 tonnes

11.2
Area of Operation of Lifting Equipment

The lifting equipment should be able to achieve the appropriate SWL as shown in Table 9 when plumbing a point one metre outboard from the ship's side over the full length of the cargo manifold including bunker and, if appropriate, vapour connections. It should also be able to provide a clear lift above the deck of at least 10 metres to any manifold connection.

11.3
Operating Criteria for Lifting Equipment

The lifting equipment fitted should be able to operate as follows:

- to top and lower under load and be fail-safe in the event of loss of power when topping or lowering;
- to slew both ways by power (not manually) under load, both when topping and lowering and when not topping and lowering;
- the hook and runner should run free without manual intervention under conditions of no load;
- to lift and lower at a rate at the hook of between 10 and 15 metres per minute under load and approximately 20 to 30 metres per minute under conditions of no load;

- to operate smoothly during lifting or lowering to a fineness of control of a maximum of one inch at the hook. The equipment should have a similar smoothness of operation when slewing. (This recommendation is particularly important for safety of operations when matching hoses to the manifold, fitting heavy reducing pieces, etc.);

- to be operated from a central control position for all functions - lifting and lowering, topping and lowering and slewing. (It is desirable that the control position be duplicated at the ship's side to enable the operator to see all stages of the operation);

- to present the hook or shackle on a single part of wire at the water's edge.
  (This recommendation is aimed to minimise the danger to personnel, either in a work boat, or ashore, or on the ship, from a large block with several turns of wire close to them. As an interim measure, a suitably-sized 3-metre pennant may be fitted under the block).

11.4
Other Criteria for Derrick Systems

All slewing operations should be powered by a single winch.

All operations-lifting and lowering, topping and lowering and slewing - should be carried out by a single winch complex.

11.5
Conflict of Criteria

Requirements of the Flag State or International regulations may be in excess of the above recommendations and should be observed in the event of conflicting criteria.

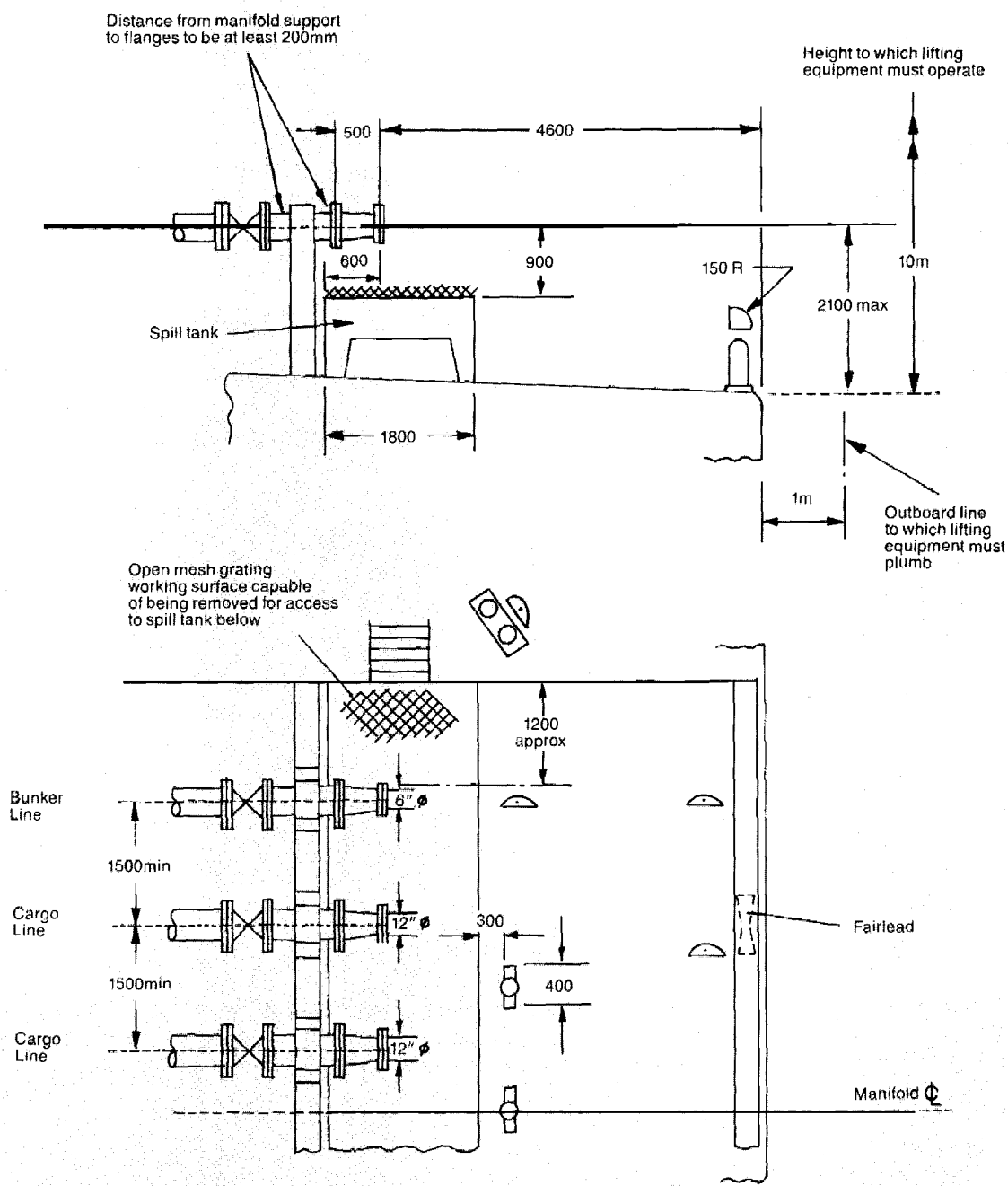


FIG. 1–CATEGORY "A"
STANDARD MANIFOLD ARRANGEMENT
FOR 16,000 TO 25,000 DWT. TANKERS

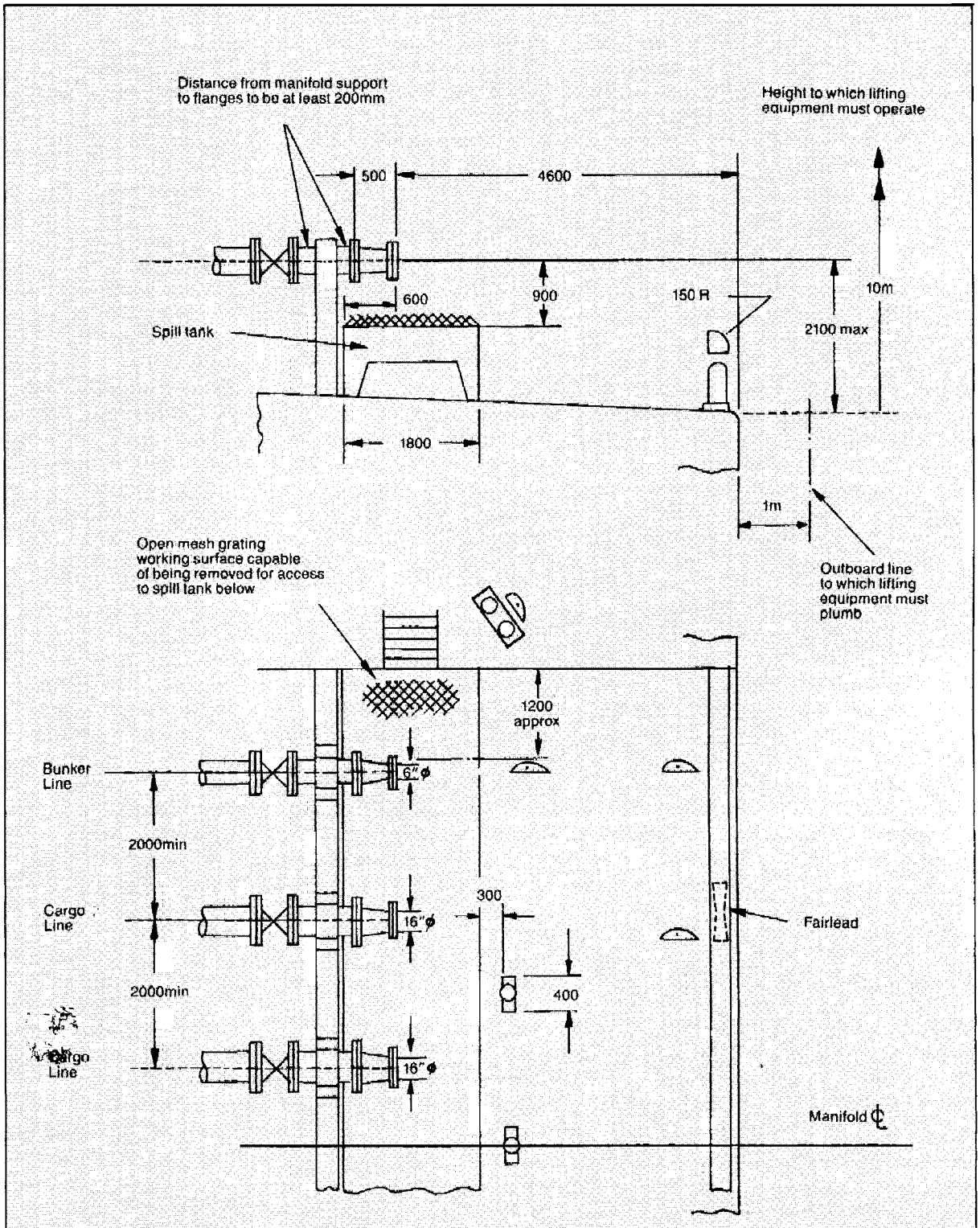


FIG. 2 – CATEGORY "B"
STANDARD MANIFOLD ARRANGEMENT
FOR 25,001 TO 60,000 DWT. TANKERS

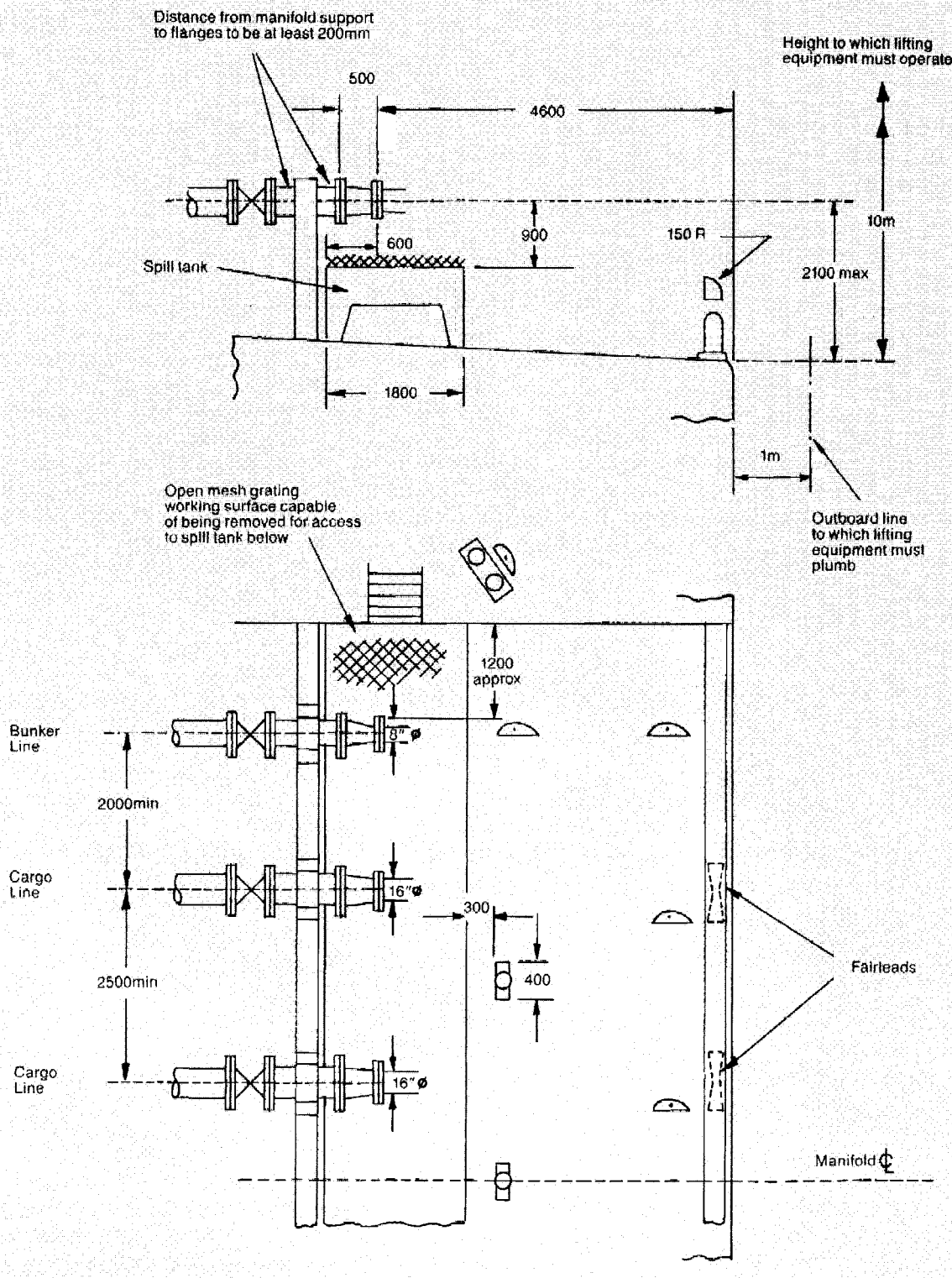


FIG. 3–CATEGORY "C"
STANDARD MANIFOLD ARRANGEMENT
FOR 60,001 TO 160,000 DWT. TANKERS

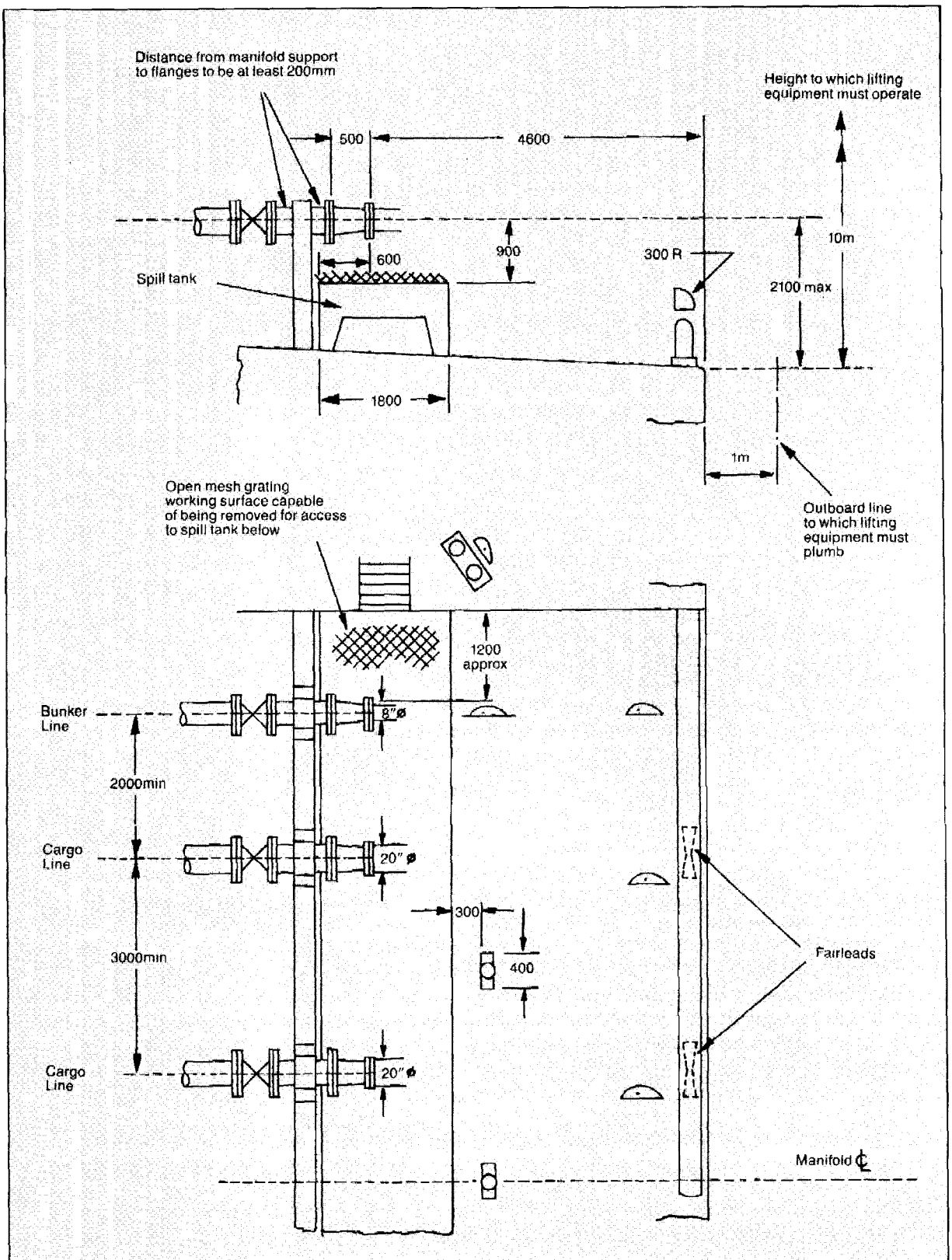
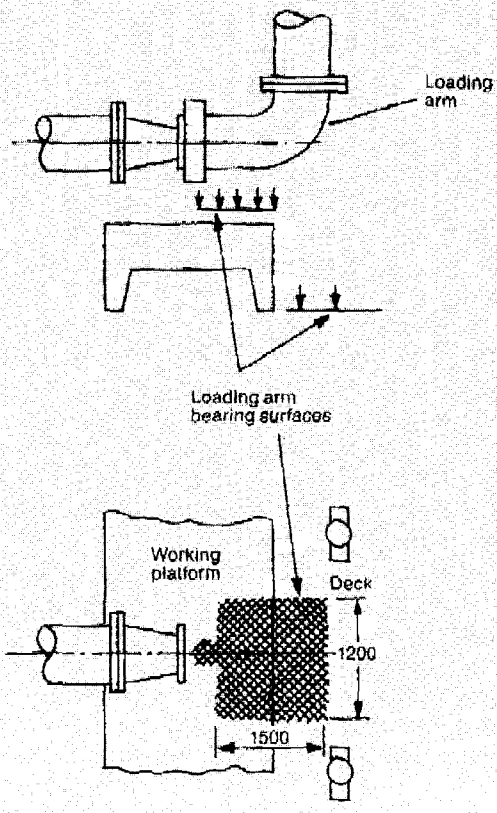
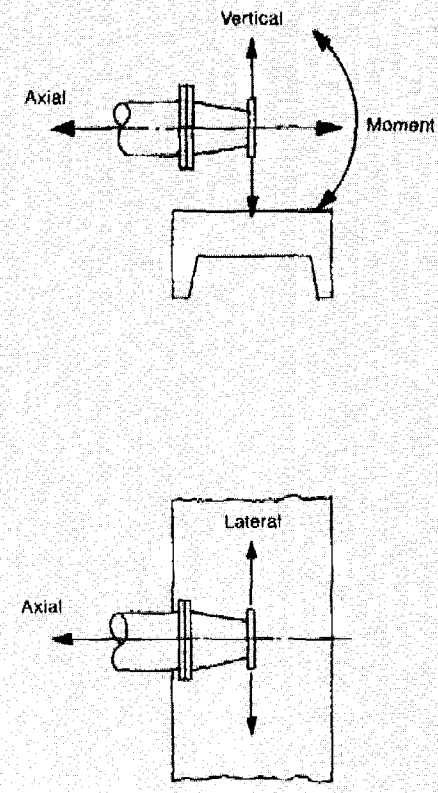


FIG. 4 – CATEGORY "D'
STANDARD MANIFOLD ARRANGEMENT
FOR TANKERS OVER 160,000 DWT



NOTE: Dimensions given are applicable to all tonnage categories

FIG. 5
LOADING ARM JACK BEARING SURFACES



NOTE: Maximum moments may be found outside of the vertical plane

FIG. 6
LOADING ARM LOADS (DEFINITION OF FORCES)

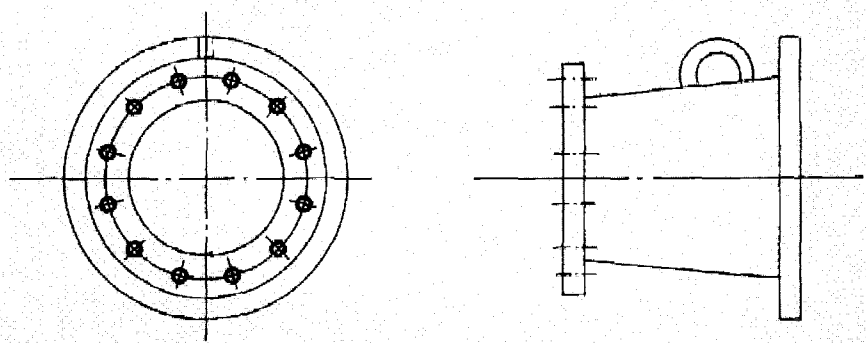


FIG. 7
PRESENTATION FLANGE ORIENTATION

Correct orientation of presentation flange bolt holes about a vertical plane

ANNEX

Recommendations for Vapour Manifolds

Note: The following recommendations are provided for the guidance of tanker operators whose ships may trade to terminals where it is a requirement for cargo vapours to be transferred ashore.

A.1 GENERAL

This Annex only provides information relating to vapour manifolds that is additional to the recommendations for cargo manifolds and associated equipment. Reference should therefore be made to the relevant section of the text for cargo manifolds for details of tonnage categories, manifold position and materials.

A.2 MANIFOLD POSITION

A.2.1 *Spacing*

The minimum spacing of the vapour manifolds as measured from centre to centre from the bunker lines along the line of presentation flanges should be in accordance with Table A.1 below.

TABLE A.1: MINIMUM VAPOUR MANIFOLD SPACING

Ship Tonnage Category	Vapour Manifold Spacing Centre to Centre
A	1.5 metres
B C D	2.0 metres

For retrofit installations aboard ships where significant obstructions prevent the location of the vapour manifolds in accordance with Table A.1, the following alternative arrangements may be considered.

a) The vapour manifolds may be located fore and aft of the bunker lines at a lesser distance than that prescribed in Table A.1. However, the distance from the bunker line should not be less than 1.0 metre.

b) Should no other arrangement be possible, the vapour manifolds may be located directly above the centrelines of the existing bunker lines ('piggy backed'). However, it should be noted that this arrangement may restrict the ship's capability to load bunkers concurrently with vapour recovery operations. If an existing working platform is to be used with this arrangement, the height of the centre of the vapour line's presentation flange above the working platform should not exceed 1.5 metres.

A.2.2 *Number and Position*

Four vapour connections should be provided, two on each side of the ship, with presentation flanges at the same height above the deck as the cargo manifold. One vapour connection should be located forward of the manifold and one located aft of the manifold on each side of the ship.

A.2.3 *Labelling*

The first 1.0 metre inboard of each manifold connection should be painted on its exterior surfaces, excluding flange faces. The painted area should be divided into three bands with the outboard and inboard bands being red in colour and 100mm wide and the centre band being yellow in colour. In addition, the word 'VAPOUR' should be painted in black letters at least 50mm high on both the forward and aft sides of all reducers, both principal and reserve, in approximately the 2 o'clock and 10 o'clock positions. (see Figure A.6)

A.3 SPILL TANK AND WORKING PLATFORM

The spill tank and working platform provided in accordance with Section 3 should extend beyond the vapour connections at the forward and aft ends of the manifold.

A.4 VAPOUR MANIFOLD:GENERAL

A.4.1
Material and Strength

The whole vapour manifold, including the manifold support, should be constructed according to the recommendations contained in Section 4.1 for cargo manifolds. The manifold supports, distance pieces and reducers should be capable of withstanding the loads at the presentation flange as detailed in Table 2.

A.5 VAPOUR MANIFOLD:SPECIFICATION AND FITTINGS

A.5.1
Valves and Distance Pieces

Valves and distance pieces should be provided in accordance with the recommendations contained in Sections 5.1 and 5.2 for cargo manifolds.

A.5.2
Reducers/Expanders

All reducers/expanders should be constructed in accordance with the recommendations contained in Section 5.3.

A.5.2.1 *Presentation Flanges*

The number of reducers/expanders and the size of presentation flanges for the various tonnage categories should be in accordance with Table A.2.

In normal circumstances, the principal vapour reducers/expanders should be kept bolted to the distance pieces. On the occasions when a smaller or larger presentation flange is required, the semi-permanent principal reducers/expanders should be removed and be replaced by the required size from the onboard stock. No more than one reducer/expander should be fitted outboard of the distance piece at any time.

TABLE A.2 VAPOUR REDUCERS/EXPANDERS

PRINCIPAL FITTINGS			RESERVE REDUCER STOCKS
Tonnage Category	Number of Reducers/ Expanders	Presentation Flange Nominal Bore	
A	4	12"	2 x 8"
B	4	12"	2 x 16"
C	4	16"	2 x 12"
D	4	16"	2 x 12"

A.5.2.2. *Material and Design*

All presentation flanges should conform to ANSI B16.5. They should be kept vertical and should have plain (flat) faces, i.e. raised faced flanges should not be used.

In order to prevent the possible misconnection of the vapour manifold to a liquid loading line, a mechanical keying arrangement should be provided for the presentation flanges as follows (see also Figure A.5):

1) The bolt locations should be arranged so that two bolts straddle the 12 o'clock position at the top of the flange face.

2) One cylindrical stud should be permanently attached to each of the presentation flange faces at the 12 o'clock position on the respective flange bolt circles. This cylindrical stud should project horizontally outboard perpendicular to the flange face and be centred on the same bolt circle as defined in ANSI B16.5. The diameter of the cylindrical stud should be 12.7mm (0.5 inches) and it should be at least 25.4mm (1.0 inches) long.

A.5.2.3.. *Types of Flanges*

Flanges should conform to the recommendations contained in Section 5.3.6.

A.5.2.4 *Blank Flanges*

Each presentation flange of the principal reducers/expanders should be provided with a removable steel blank flange fitted with handles. Each blank flange should be painted yellow on its exteriour surfaces (excluding the inboard – facing flange face). To accommodate the lug on the vapour line presentation flange, each blank flange should have an extra hole located in the 12 o'clock position on the flange's bolt hole circle. The diameter of the hole should be 15.88 mm (0.625 inches).

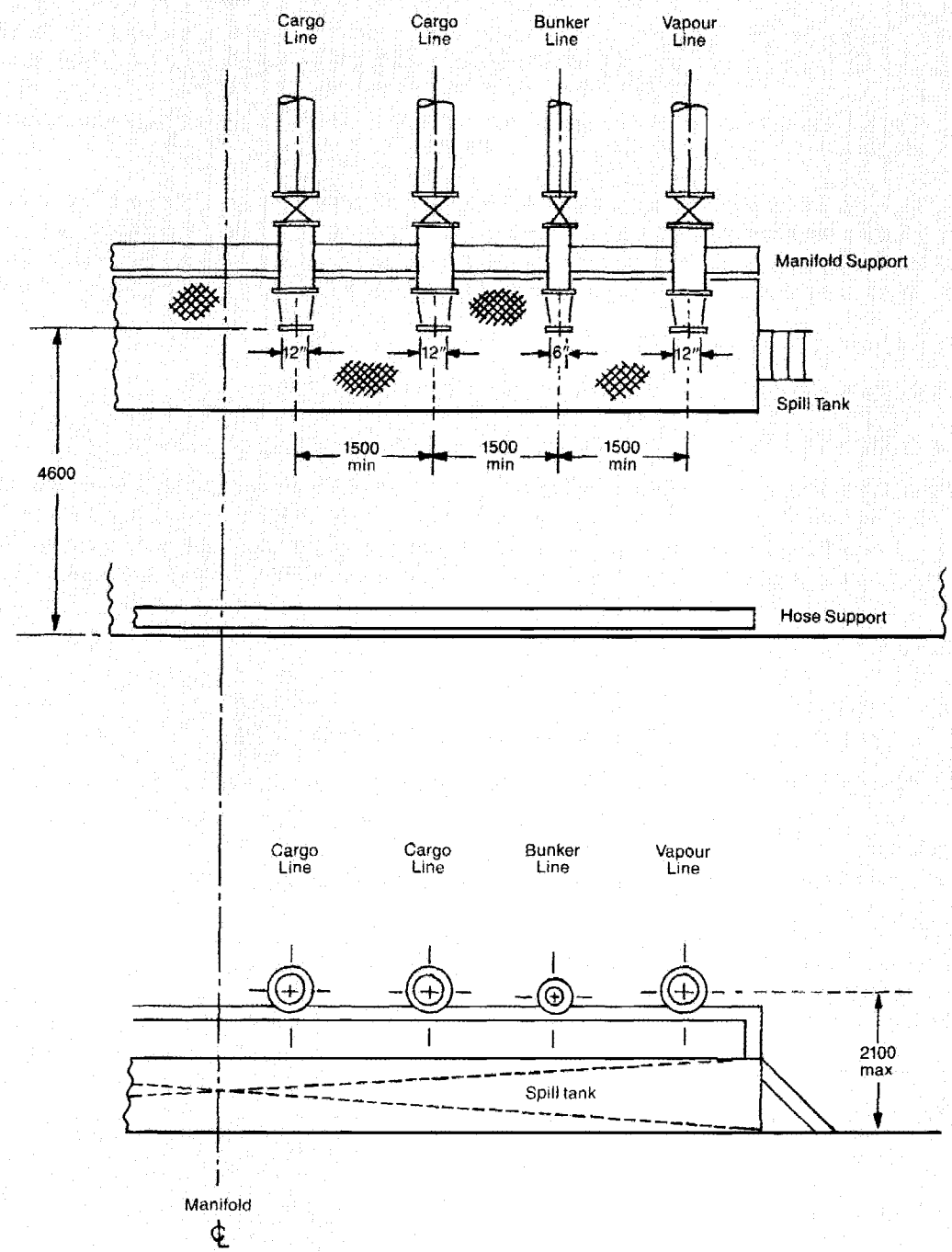


FIG. A.1 — CATEGORY "A"
STANDARD MANIFOLD ARRANGEMENT FOR 16,000 TO 25,000 DWT TANKERS INCLUDING VAPOUR LINES

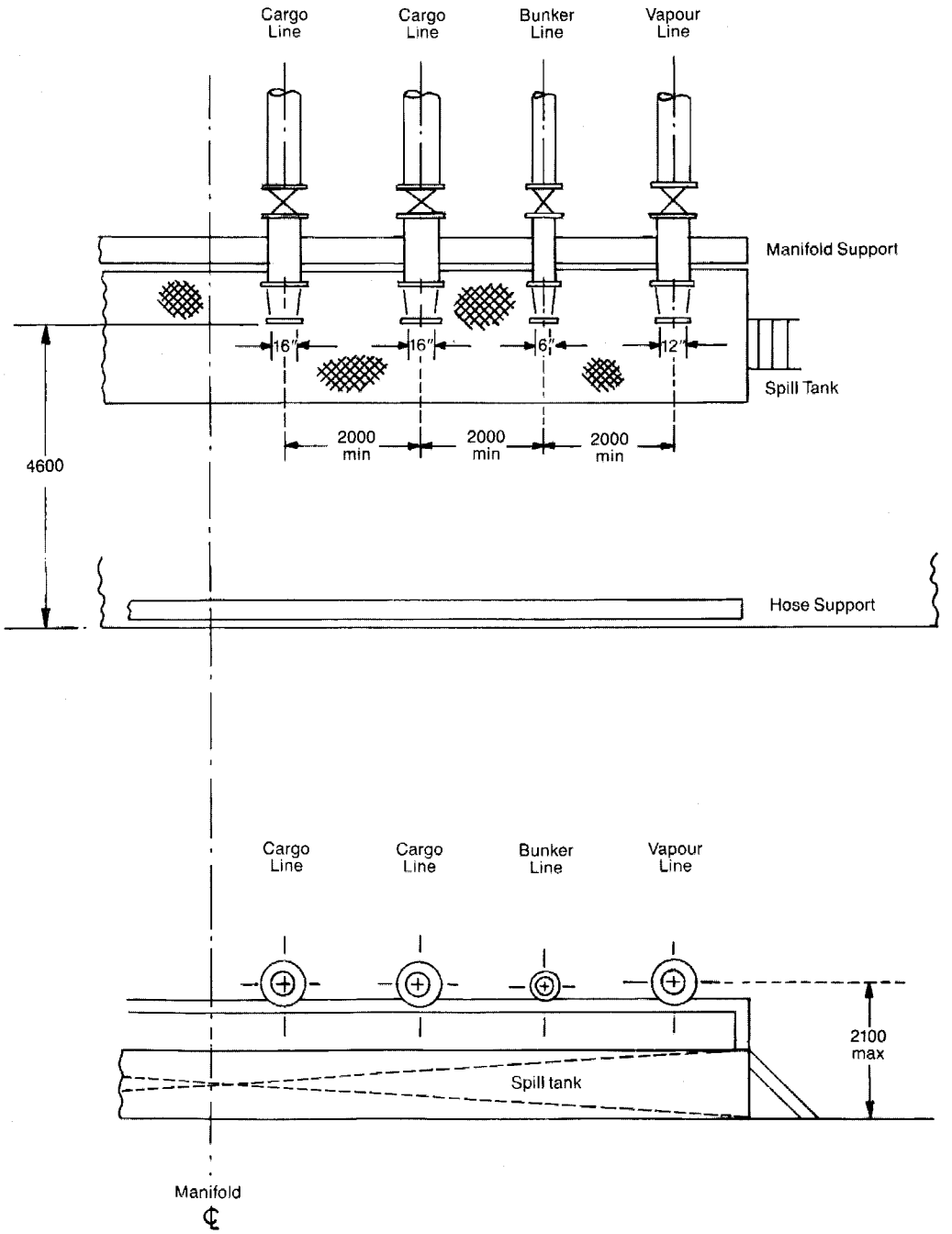


FIG. A.2 — CATEGORY "B"
STANDARD MANIFOLD ARRANGEMENT FOR 25,001 TO 60,000 DWT TANKERS INCLUDING VAPOUR LINES

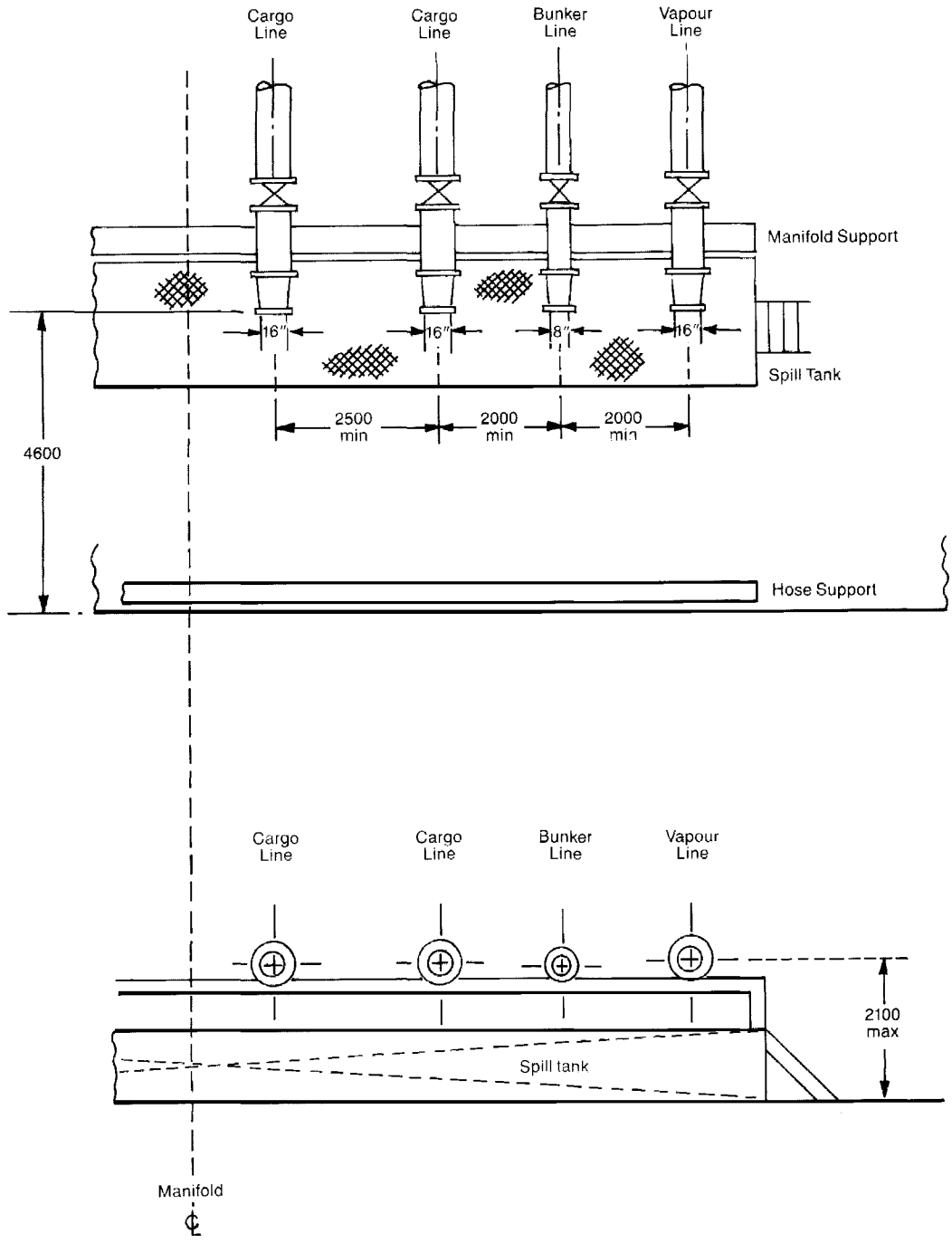


FIG. A.3 — CATEGORY "C"
STANDARD MANIFOLD ARRANGEMENT FOR 60,001 TO 160,000 DWT TANKERS INCLUDING VAPOUR LINES

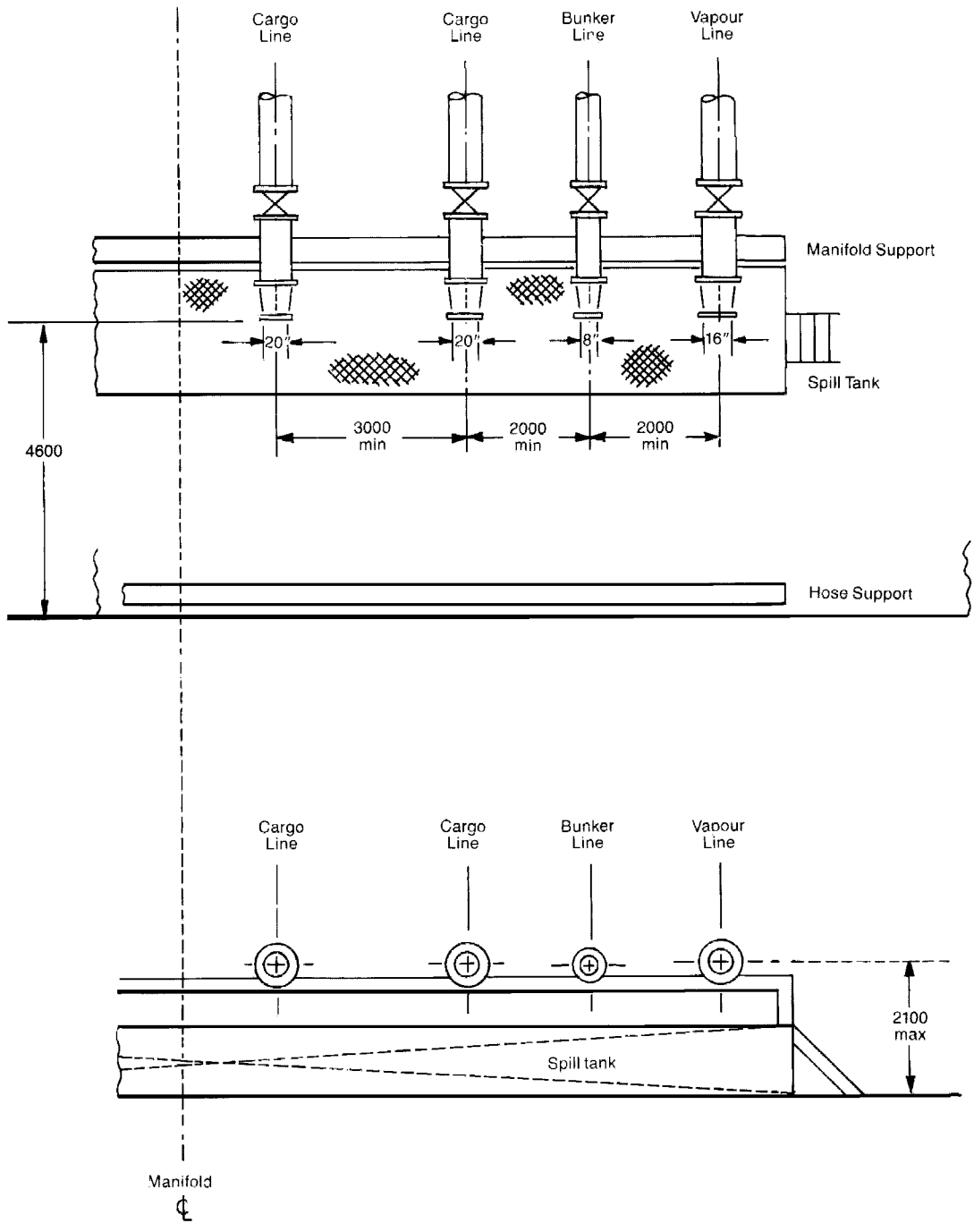


FIG. A.4 — CATEGORY "D"
STANDARD MANIFOLD ARRANGEMENT FOR TANKERS
OVER 160,000 DWT INCLUDING VAPOUR LINES

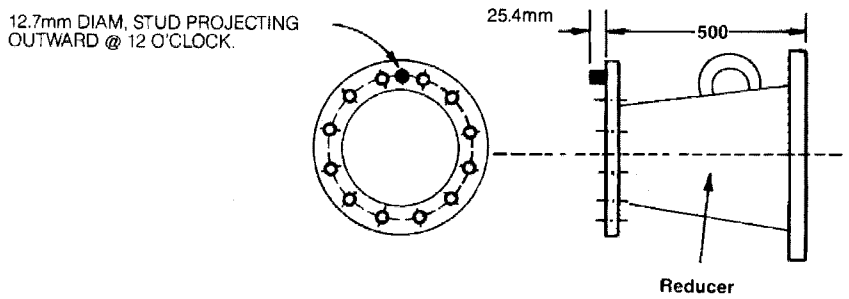


FIG. A.5
VAPOUR MANIFOLD PRESENTATION FLANGE ORIENTATION

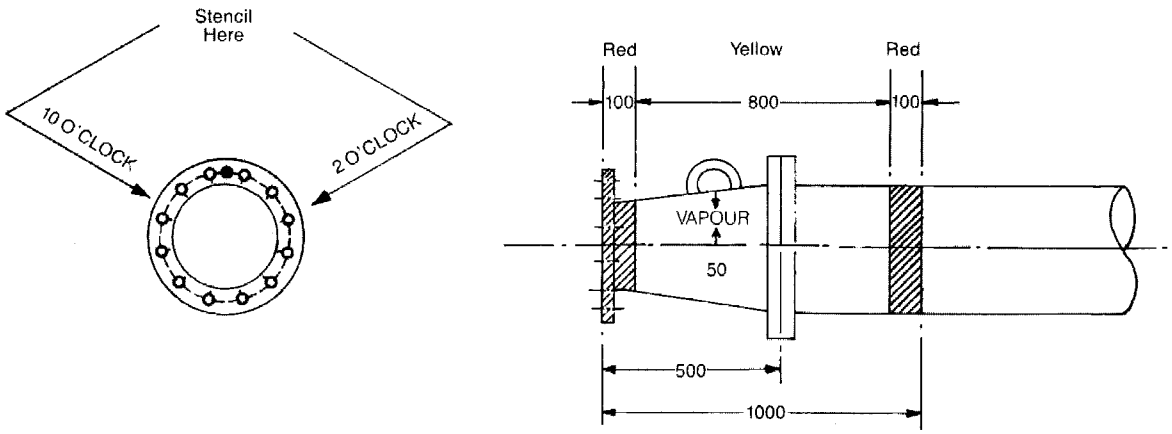


FIG. A.6
LABELLING OF VAPOUR MANIFOLDS